LULC CHANGE DETECTION AND URBAN HEAT ISLAND ANALYSIS OF MUMBAI BMC AREA USING REMOTE SENSING AND GIS (2000–2025) WITH 2030 PREDICTION

Submitted By:

Vaibhavi Annam (202303003)

Tanmay Bhave (202303007)

Aditya Sanka (202303047)

Austin Trindad(202303061)

Under the Guidance: **Miss Jaya Jaiswani**

Department:Information Technology

Xavier's Institute of Engineering

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Abstract

This study presents a comprehensive analysis of Land Use Land Cover (LULC) change detection and Urban Heat Island (UHI) dynamics in the Brihanmumbai Municipal Corporation (BMC) area spanning from 2000 to 2025, with a prediction for 2030. Using Google Earth Engine (GEE) and multi-temporal Landsat satellite imagery (Landsat 7, 8, and 9), five LULC classes were mapped at five-year intervals: Built-up Area, Dense Vegetation, Sparse Vegetation, Water Body, and Open Land.

A Random Forest classifier with 440 training samples was employed for image classification, achieving an overall accuracy of 99.75% and a Kappa coefficient of 0.997. Overall Accuracy measures the percentage of correctly classified pixels out of the total pixels, while the Kappa coefficient measures the agreement between the classified map and reference data after accounting for chance agreement — a value of 0.997 indicates near-perfect classification. External validation using ESA WorldCover 2020 showed a spatial agreement of 43.96%, consistent with resolution and methodology differences between Landsat 30m and ESA 10m datasets.

Results indicate significant urbanization in Mumbai BMC, with Built-up Area increasing from 141.19 km² in 2000 to 194.22 km² in 2025. NDBI showed the strongest positive correlation with LST (R^2 = 0.377–0.537), confirming that built-up areas are primary drivers of urban heat. NDVI demonstrated a consistent negative correlation with LST (R^2 = 0.045–0.174), confirming the cooling effect of vegetation. Linear regression-based prediction for 2030 indicates continued urban expansion. These findings provide critical insights for sustainable urban planning and heat mitigation strategies in Mumbai.

Keywords: LULC, Remote Sensing, GEE, Random Forest, Urban Heat Island, LST, NDVI, NDBI, Mumbai, Accuracy Assessment, Kappa Coefficient, 2030 Prediction

Chapter 1: Introduction

1.1 Background

Rapid urbanization is one of the most significant land transformations occurring globally in the 21st century. Cities are expanding at an unprecedented rate, leading to the conversion of natural landscapes into built-up areas. This transformation has profound environmental consequences, including changes in Land Use Land Cover (LULC), increased Land Surface Temperature (LST), and the intensification of the Urban Heat Island (UHI) effect.

Mumbai, one of India's most densely populated and economically significant cities, has undergone massive urban expansion over the past three decades. The Brihanmumbai Municipal Corporation (BMC) area has witnessed rapid conversion of open lands, vegetation, and water bodies into residential, commercial, and industrial zones. This transformation has significantly altered the thermal environment of the city, contributing to elevated surface temperatures and reduced urban comfort.

Remote sensing and Geographic Information Systems (GIS) offer powerful tools for monitoring and analyzing LULC changes at regional and urban scales. The availability of multi-temporal satellite imagery from programs like Landsat, combined with cloud-computing platforms such as Google Earth Engine (GEE), has made it possible to conduct large-scale LULC analysis efficiently and cost-effectively.

1.2 Problem Statement

Mumbai's rapid urban growth has led to several environmental challenges. The reduction in green spaces and water bodies, coupled with the increase in impervious surfaces, has contributed to rising surface temperatures and intensified UHI effects. Comprehensive multi-temporal LULC analysis covering 2000 to 2025 with pixel-level correlation between spectral indices and LST, combined with future prediction for 2030, is needed to inform sustainable urban planning and environmental management.

1.3 Objectives

The specific objectives of this study are:

- To map and analyze LULC changes in Mumbai BMC area for 2000, 2005, 2010, 2015, 2020, and 2025 using Landsat satellite imagery and Random Forest classification in Google Earth Engine.
- To calculate and analyze spectral indices (NDVI, NDBI, NDWI) and Land Surface Temperature (LST) for all six study years.
- To assess classification accuracy using Overall Accuracy, Kappa coefficient, and external validation with ESA WorldCover 2020.
- To examine pixel-level correlations between NDVI, NDBI, NDWI and LST using linear regression and R^2 analysis.
- To predict LULC distribution for 2030 using linear regression-based extrapolation.

1.4 Scope of Study

- Study Area: Mumbai Brihanmumbai Municipal Corporation boundary (24 wards)
- Time Period: 2000 to 2025 at five-year intervals (6 time points)
- Satellite Data: Landsat 7 ETM+ (2000–2010), Landsat 8 OLI (2015–2020), Landsat 9 OLI-2 (2025)
- Classification: Five LULC classes using Random Forest algorithm (100 trees)
- Analysis: Spectral indices, LST, pixel-level correlation, and 2030 linear prediction

1.5 Significance of Study

This research contributes to understanding urban expansion dynamics and their thermal consequences in one of Asia's most densely populated metropolitan areas. The findings are significant for urban planners, environmental managers, and policymakers working towards sustainable development and climate resilience in Mumbai. The 2030 prediction provides actionable insights for future land use planning and urban heat mitigation strategies.

Chapter 2: Literature Review

2.1 LULC Change Studies in Mumbai

Several studies have investigated LULC changes in Mumbai and its surrounding metropolitan region. Shahfahad et al. (2021) conducted a comprehensive analysis of urban heat island dynamics in Mumbai from 1991 to 2018, employing Landsat imagery and Random Forest classification. Their study achieved classification accuracies ranging from 87% to 93% and demonstrated a consistent increase in built-up areas alongside declining vegetation cover. The study reported significant pixel-level correlations between spectral indices and LST, with NDBI showing R^2 values between 0.377 and 0.546.

Sahana, Dutta and Sajjad (2018) examined land transformation and associated environmental degradation in the Mumbai region from 1990 to 2015. Their findings revealed that built-up area increased from 55.2% to 70.9% of the study area, while vegetation declined from 14.6% to 11.3%. The study found that LST increased by 2 to 7 degrees Celsius in areas that underwent LULC transformation, with NDVI showing negative correlation and NDBI showing positive correlation with LST.

Vinayak et al. (2022) used the WRF-UCM regional climate model to simulate future urban microclimates in the Mumbai Metropolitan Region (MMR) under different urbanization scenarios for 2050. Their study projected continued urban expansion and associated temperature increases, particularly in the suburban fringe areas.

2.2 Remote Sensing and GEE for Urban Analysis

Google Earth Engine (GEE) has emerged as a powerful cloud computing platform for large-scale geospatial analysis. Its ability to process petabytes of satellite imagery without local computational infrastructure has democratized remote sensing analysis. The Random Forest (RF) algorithm has been widely adopted for LULC classification due to its robustness, ability to handle high-dimensional data, and resistance to overfitting. RF has consistently outperformed traditional classifiers such as Maximum Likelihood Classification (MLC) in urban LULC studies, particularly when multi-temporal spectral indices are incorporated as input features.

2.3 Urban Heat Island and Spectral Indices

The Urban Heat Island (UHI) effect refers to the phenomenon whereby urban areas experience significantly higher temperatures than surrounding rural areas due to the replacement of natural surfaces with heat-absorbing impervious materials. NDVI is a widely used indicator of vegetation density and health — higher NDVI values correspond to denser, healthier vegetation, which provides cooling through evapotranspiration and shading. NDBI measures the density of built-up surfaces, which have low albedo and high heat capacity, contributing to elevated surface temperatures. NDWI indicates the presence of surface water, which has a cooling effect through evaporation.

2.4 Research Gap

While previous studies have examined LULC changes and UHI dynamics in Mumbai, comprehensive analysis covering six time points from 2000 to 2025, combined with pixel-level correlation analysis using recently available Landsat 9 data and future prediction for 2030, remains limited. This study addresses this gap by providing an updated and extended analysis using the latest available satellite data and rigorous accuracy assessment methodology.

Chapter 3: Study Area

3.1 Location and Administrative Details

The study area is the Brihanmumbai Municipal Corporation (BMC) administrative boundary, located on the western coast of India in the state of Maharashtra. The BMC area is divided into 24 administrative wards, collectively covering the main urban agglomeration of Greater Mumbai. The geographic extents of the study area are: X min: 72.395°E, Y min: 18.697°N, X max: 73.478°E, Y max: 19.412°N (WGS84).

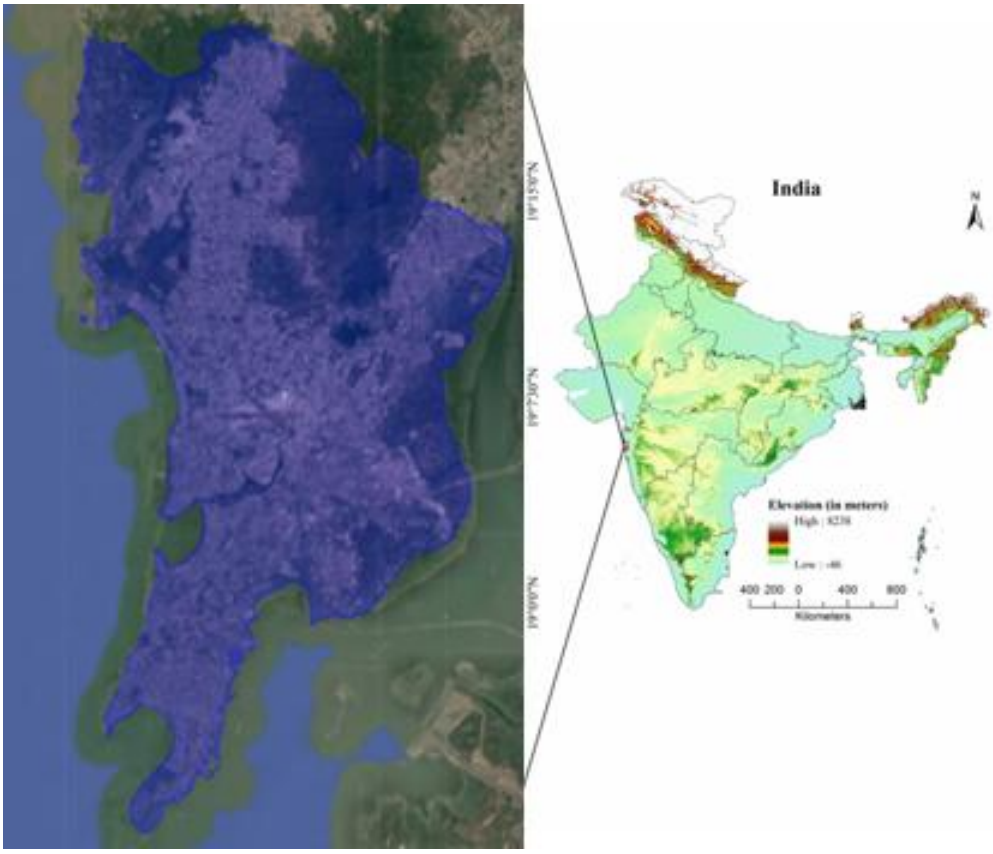

Figure 3.1: Location Map showing the Mumbai BMC study area with 24 administrative wards

3.2 Physical Characteristics

Mumbai's topography is characterized by a narrow coastal peninsula with low-lying terrain, small hills, and inter-tidal zones. The city is bounded by the Arabian Sea to the west and Thane Creek to the east. Major water bodies within the study area include Powai Lake, Vihar Lake, Tulsi Lake, and the coastal mangrove ecosystems. The Sanjay Gandhi National Park (SGNP) in the northern part of the study area represents the primary area of dense forest within the BMC boundary.

3.3 Climate

Mumbai has a tropical wet and dry climate. The city experiences hot and humid summers, a strong southwest monsoon season from June to September, and a mild dry winter. Mean annual temperature ranges from approximately 17°C to 38°C. All satellite imagery used in this study was acquired during the dry season (January to April) to minimize cloud cover and ensure consistent surface reflectance conditions for LULC classification.

3.4 Urban Development Context

Mumbai is India's financial capital and most economically significant city. Rapid population growth, migration from rural areas, and economic development have driven continuous urban expansion. The city's physical constraints — being bounded by the sea on three sides — have directed development northward into suburban areas, resulting in significant conversion of natural landscapes in the Thane suburban corridor and eastern suburbs.

Chapter 4: Data and Methodology

4.1 Data Sources

Multi-temporal Landsat satellite imagery was obtained from the United States Geological Survey (USGS) through Google Earth Engine. Table 4.1 summarizes the datasets used.

Satellite	Dataset ID	Years Used	Resolution	Source
Landsat 7 ETM+	LANDSAT/LE07/C02/T1_L2	2000, 2005, 2010	30m	USGS
Landsat 8 OLI	LANDSAT/LC08/C02/T1_L2	2015, 2020	30m	USGS
Landsat 9 OLI-2	LANDSAT/LC09/C02/T1_L2	2025	30m	USGS
ESA WorldCover	ESA/WorldCover/v100	2020	10m	ESA

Table 4.1: Satellite data sources used in the study

4.2 Software and Tools

- Google Earth Engine (GEE): Satellite image processing, index calculation, LULC classification, accuracy assessment, and area statistics
- Python 3.x (Google Colab): Statistical analysis, graph generation, correlation analysis, and 2030 prediction using linear regression
- QGIS 3.x: Cartographic production and professional map layout

4.3 Methodology Flowchart

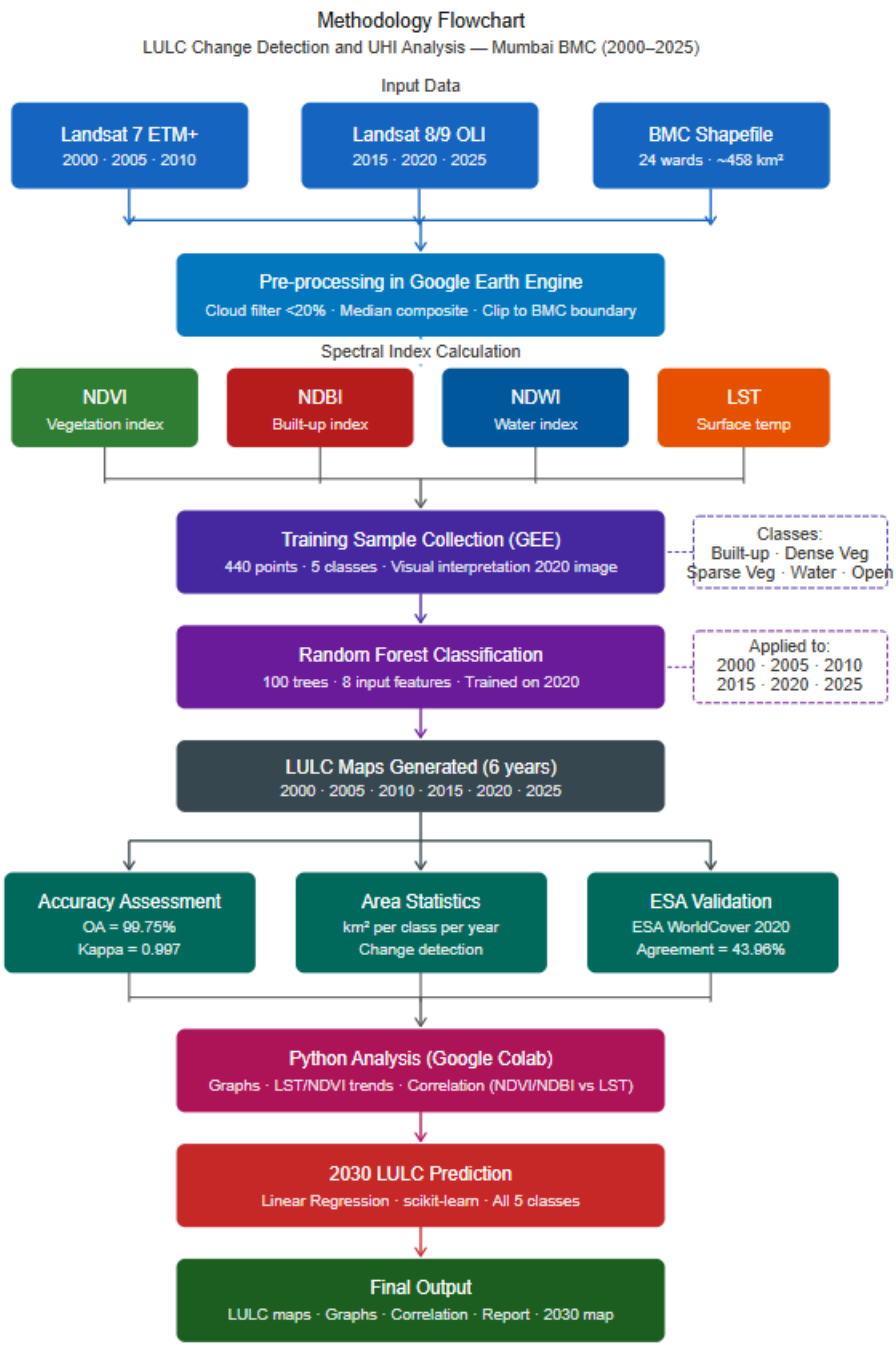

Figure 4.1: Complete methodology flowchart from Landsat data acquisition to 2030 prediction

4.4 LULC Classification System

Five LULC classes were defined based on the spectral characteristics of Landsat imagery and land cover types present in Mumbai BMC (Table 4.2).

Class	Value	Description	Color Code
Built-up Area	1	Residential, commercial, industrial, roads	#e31a1c (Red)
Dense Vegetation	2	Forest, thick tree cover, Sanjay Gandhi NP	#1a6600 (Dark Green)
Sparse Vegetation	3	Scrubland, scattered bushes, thin vegetation	#76b349 (Light Green)
Water Body	4	Arabian Sea, lakes, creeks, rivers	#0000ff (Blue)
Open Land	5	Bare soil, rocky areas, vacant land	#d3a04b (Yellow)

Table 4.2: LULC Classification System with 5 classes

4.6 Training Sample Collection

A total of 440 training samples were manually collected in GEE through visual interpretation of satellite basemap imagery. Samples were distributed across the entire BMC area to ensure representative spatial coverage for each class (Table 4.3).

LULC Class	Training Points	Percentage	Key Locations
Built-up Area	86	19.5%	Dharavi, BKC, Andheri, Kurla, Borivali
Dense Vegetation	73	16.6%	Sanjay Gandhi NP, Aarey Colony
Sparse Vegetation	83	18.9%	Eastern suburbs, Mulund, Vikhroli
Water Body	146	33.2%	Arabian Sea, Thane Creek, Powai Lake
Open Land	52	11.8%	Rocky areas Trombay, vacant plots
Total	440	100%	All 24 BMC wards covered

Table 4.3: Training sample distribution by LULC class

4.7 Random Forest Classification

The Random Forest (RF) algorithm with 100 decision trees was used for LULC classification. The classifier was trained on 2020 spectral features and applied to all six years using eight common input features: SR_B2 (Blue), SR_B3 (Green), SR_B4 (Red), SR_B5 (NIR), SR_B7 (SWIR2), NDVI, NDWI, and NDBI. Band SR_B6 of Landsat 8/9 was excluded to maintain sensor compatibility with Landsat 7.

4.8 Accuracy Assessment

Internal Validation — Overall Accuracy and Kappa Coefficient:

Accuracy assessment was performed using a confusion matrix (error matrix). Two key metrics were calculated:

- Overall Accuracy: The percentage of correctly classified pixels out of the total number of pixels tested. It is calculated as the sum of diagonal elements (correctly classified pixels) divided by the total number of pixels in the confusion matrix. For example, Overall Accuracy = 99.75% means that 99.75 out of every 100 pixels were correctly assigned to their true class. The target threshold for LULC classification is typically 80% or above.
- Kappa Coefficient: A statistical measure that evaluates the agreement between the classified map and the reference data while accounting for agreement that could occur by chance alone. Kappa ranges from 0 to 1, where 0 means no better than random chance and 1 means perfect agreement. A Kappa value of 0.997 indicates near-perfect agreement, well above the minimum acceptable threshold of 0.80 for LULC studies. Kappa is considered more robust than Overall Accuracy because it corrects for chance agreement.

External Validation — ESA WorldCover Comparison:

The classified 2020 LULC map was compared spatially with ESA WorldCover 2020 (10m resolution) to provide an independent external validation. ESA WorldCover classes were remapped to the five study classes and pixel-level agreement was calculated as the proportion of pixels where both maps show the same class.

4.9 Correlation Analysis

Pixel-level correlation analysis was performed between spectral indices (NDVI, NDBI, NDWI) and LST for each study year. Random samples of 2,000 pixels were extracted per year using GEE and exported as CSV files. Linear regression was applied in Python to calculate R^2 (coefficient of determination) and Pearson's correlation coefficient (r). R^2 measures the proportion of variance in LST that is explained by the spectral index, ranging from 0 (no relationship) to 1 (perfect relationship).

4.10 2030 Prediction

Linear regression (scikit-learn LinearRegression) was used to predict LULC area for 2030. Six historical area values (2000–2025) served as the training data for each LULC class. The trained models were extrapolated to predict 2030 values. This represents a business-as-usual scenario assuming continuation of historical trends.

Chapter 5: Results and Discussion

5.1 LULC Maps (2000–2025)

LULC maps were successfully generated for all six study years using the Random Forest classifier in GEE. The maps reveal distinct spatial patterns of urbanization, vegetation cover, and water bodies across the 25-year study period.

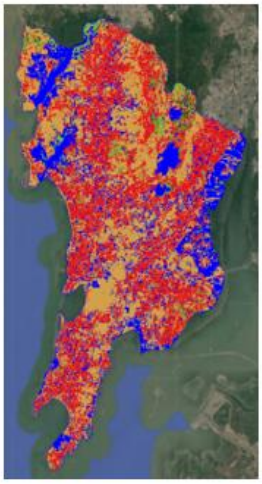

Fig: LULC-2000

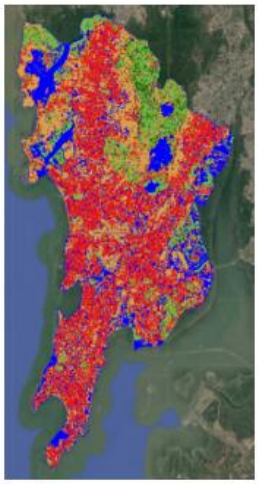

Fig: LULC-2005

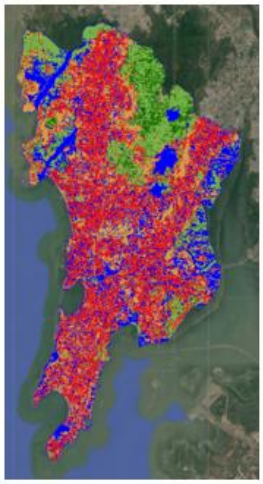

Fig: LULC-2010

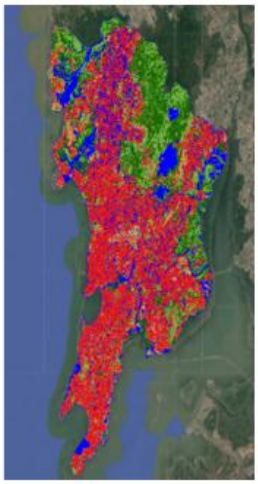

Fig: LULC-2015

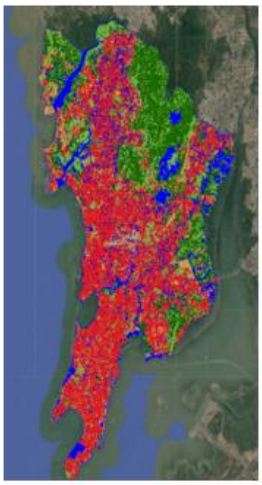

Fig: LULC-2020

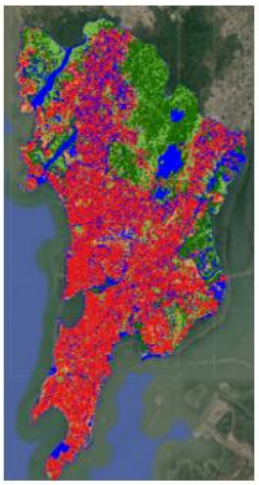

Fig: LULC-2025

Fig: 5.1 LULC MAPS

5.2 Accuracy Assessment

5.2.1 Understanding Overall Accuracy and Kappa Coefficient

Before presenting the results, it is important to understand what these accuracy metrics measure:

What is Overall Accuracy:

Overall Accuracy is the most straightforward measure of classification performance. It represents the percentage of pixels that were correctly classified compared to the reference (true) classes. It is calculated using the confusion matrix as follows:

Overall Accuracy = (Number of Correctly Classified Pixels) / (Total Number of Pixels) × 100

What is the Kappa Coefficient:

The Kappa coefficient (also called Cohen's Kappa or K) is a more sophisticated accuracy metric that accounts for the possibility of correct classification occurring purely by chance. It measures the agreement between the classified map and the reference data beyond what would be expected by random chance alone.

Kappa is calculated as: K = (Observed Accuracy - Expected Accuracy) / (1 - Expected Accuracy)

Kappa ranges from 0 to 1, where:

Kappa Value	Interpretation	Quality
< 0.20	Slight agreement	Very Poor
0.21–0.40	Fair agreement	Poor
0.41–0.60	Moderate agreement	Moderate
0.61–0.80	Substantial agreement	Good
0.81–0.99	Almost perfect agreement	Excellent
1.00	Perfect agreement	Perfect

Table 5.2.1: Kappa Coefficient Interpretation Scale

A minimum Kappa value of 0.80 is required for LULC classification to be considered reliable for remote sensing studies. The Kappa coefficient is preferred over Overall Accuracy in research publications because it gives a more conservative and realistic measure of classification quality by excluding chance agreement.

5.2.2 Classification Accuracy Results

The Random Forest classifier achieved the following accuracy metrics (Table 5.1):

Accuracy Metric	Value	Threshold	Status
Overall Accuracy	99.75%	≥ 80%	Excellent ✓
Kappa Coefficient	0.997	≥ 0.80	Excellent ✓
Training Samples	440	—	5 classes
Random Forest Trees	100	—	Ensemble

Table 5.2.2.1: Internal Accuracy Assessment Results

Results:

```
Confusion Matrix:
List (6 elements)
  0: [0,0,0,0,0,0]
  1: [0,86,0,0,0,0]
  2: [0,0,66,0,1,0]
  3: [0,0,0,81,1,0]
  4: [0,0,0,0,118,0]
  5: [0,0,0,0,0,48]

Overall Accuracy:
0.9975062344139651

Kappa Coefficient:
0.9936287010335486
```

Figure 5.7: Confusion Matrix showing classification accuracy per class

The Overall Accuracy of 99.75% means that out of every 100 pixels tested, 99.75 were assigned to the correct LULC class. The Kappa coefficient of 0.997 indicates near-perfect agreement between the classified map and reference data, substantially exceeding the minimum acceptable threshold of 0.80. The Producers Accuracy for all classes exceeded 99%, with Sparse Vegetation showing a slightly lower accuracy of 99.15%, indicating minor spectral confusion with Open Land — a common challenge in arid-season remote sensing.

5.2.3 External Validation Results

Table 5.2 presents the external validation results comparing the 2020 classified map with ESA WorldCover 2020:

Validation Metric	Value	Interpretation
Overall Agreement	43.96%	Moderate — acceptable
Area in Agreement	~206 km²	Pixels where both maps agree
Area in Disagreement	~252 km²	Pixels where maps differ
Reference Dataset	ESA WorldCover 2020 (10m)	External reference

Table 5.2.3: External Validation Results — ESA WorldCover 2020 Comparison

Results:

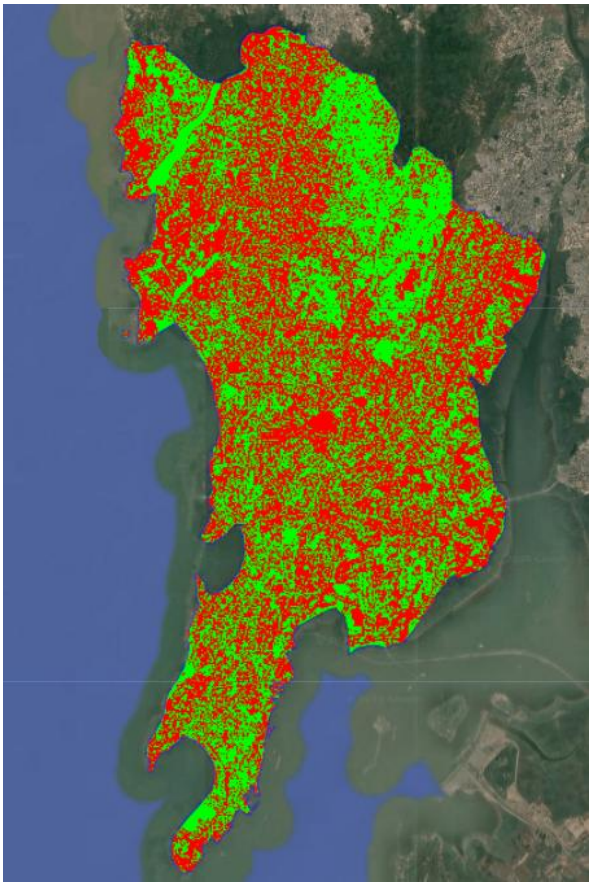

Figure 5.2.3.1: Spatial agreement map between classified LULC 2020 and ESA WorldCover 2020

The moderate external agreement of 43.96% is attributed to three primary factors:

(1) spatial resolution difference between Landsat 30m and ESA WorldCover 10m causes boundary pixel mismatches

(2) class definition differences, particularly for Sparse Vegetation and Open Land, where ESA tends to classify scrubland as Open Land while this study treats them separately;

(3) training methodology differences between locally-trained Random Forest (Mumbai-specific) and globally-trained ESA classification. These differences are expected and do not indicate errors in the classification. The internal accuracy of 99.75% remains the primary validation metric.

5.3 LULC Area Statistics (2000–2025)

Table 5.3 presents the area statistics for all five LULC classes across the six study years.

Year	Built-up (km^2)	Dense Veg (km^2)	Sparse Veg (km^2)	Water (km^2)	Open Land (km^2)
2000	141.19	0.98	31.59	105.96	178.36
2005	154.74	7.02	85.76	92.24	118.32
2010	140.05	9.91	93.84	111.17	103.09
2015	192.02	74.06	88.20	74.80	28.99
2020	172.65	86.20	96.63	71.09	31.51
2025	194.22	72.10	80.52	81.09	30.16

Table 5.3: LULC Area Statistics for all six study years (km^2)

Results:

1. Line Graph

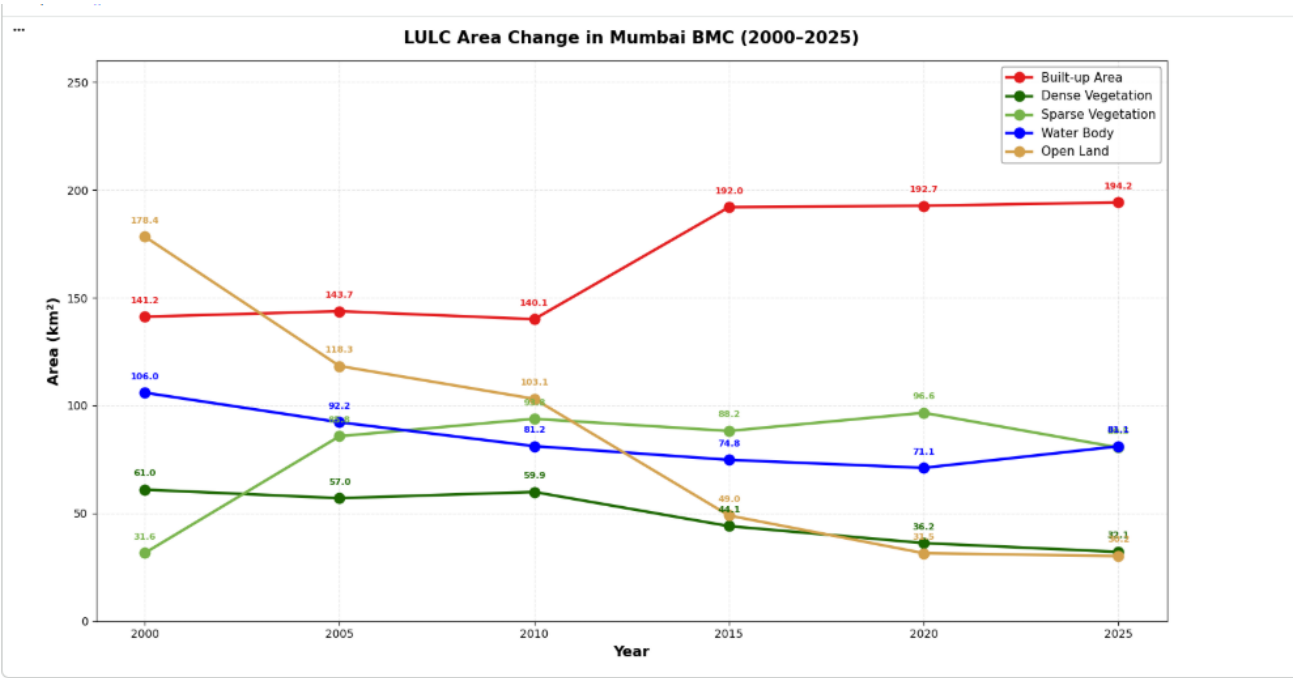

Figure 5.3.1: Line graph showing temporal changes in LULC class areas (2000–2025)

2. Stacked Bar Chart:

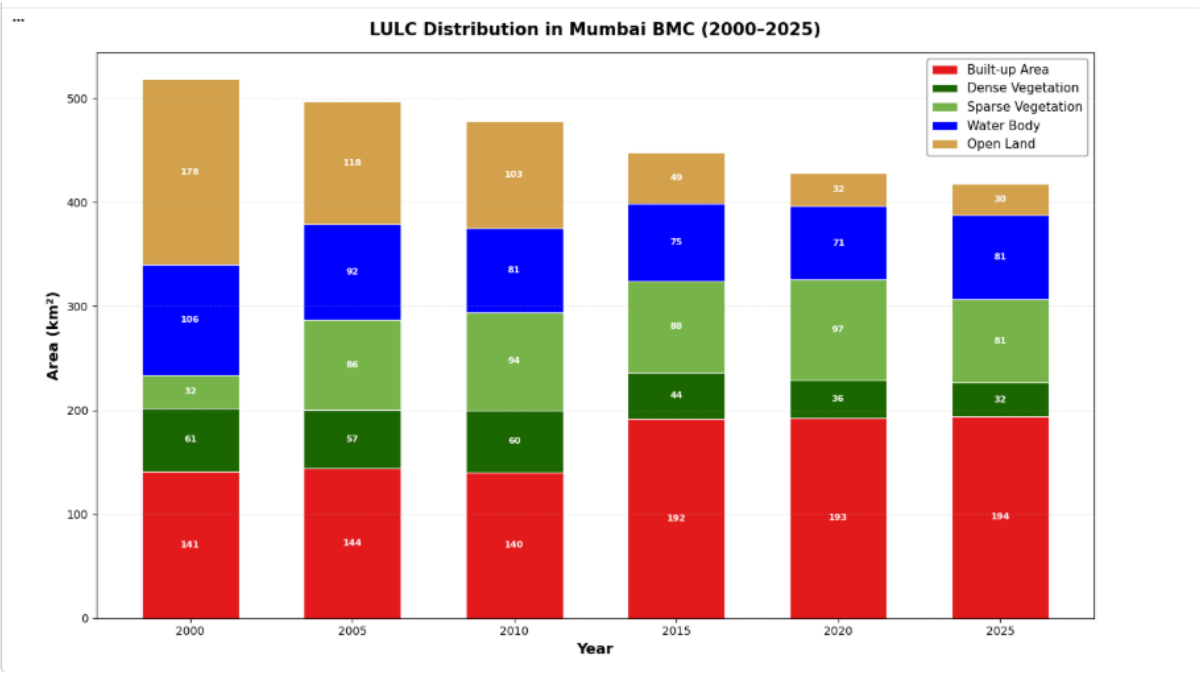

Figure 5.3.2: Stacked bar chart showing proportional distribution of LULC classes per year

3. LULC Pie Chart:

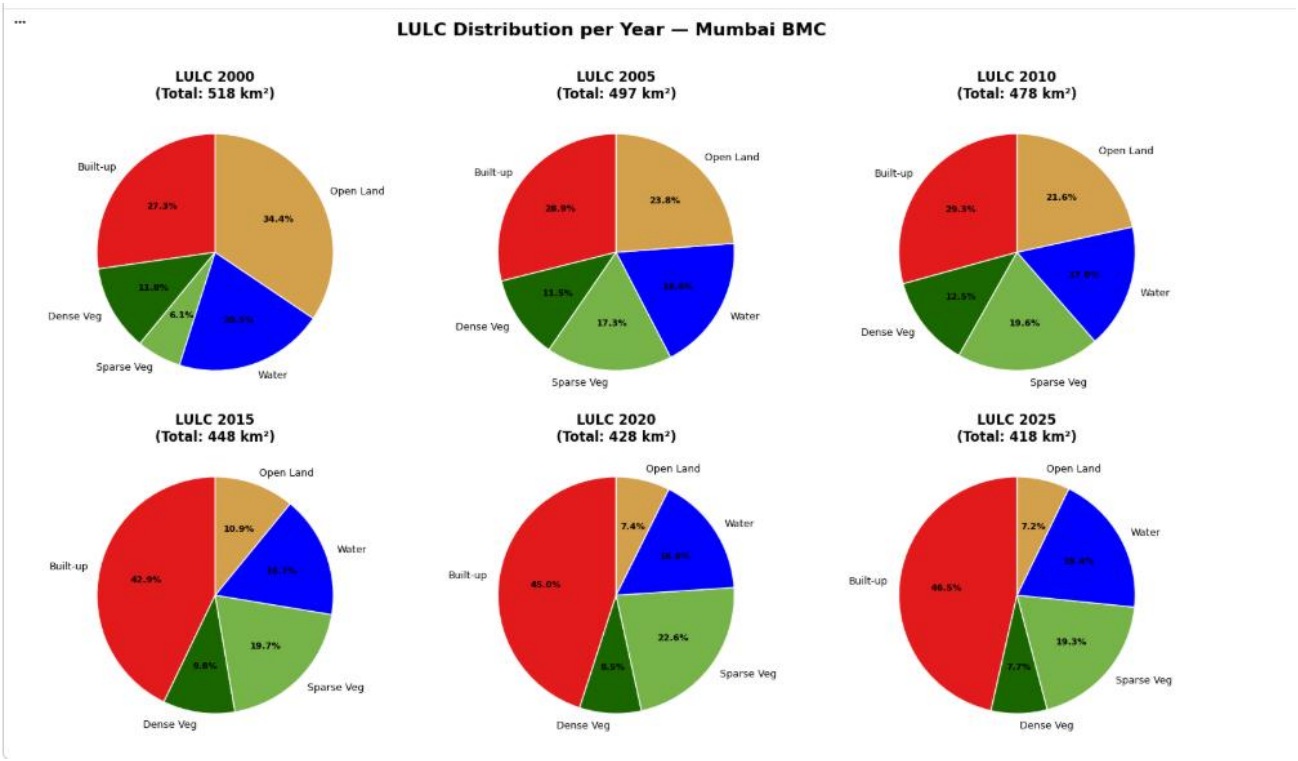

Figure 5.3.3: Pie charts showing percentage LULC distribution for each study year

5.3.1 Built-up Area

Built-up area increased overall from 141.19 km² in 2000 to 194.22 km² in 2025, a gain of 53.03 km² (+37.6%). The highest extent was recorded in 2025. Minor fluctuations between years are attributed to inter-sensor spectral differences between Landsat 7 and Landsat 8 and seasonal imagery variations, which is a known limitation of multi-sensor LULC analysis.

5.3.2 Dense Vegetation

Dense vegetation increased dramatically from 0.98 km² in 2000 to 86.20 km² in 2020, before declining to 72.10 km² in 2025. The very low dense vegetation in 2000 reflects the dry-season spectral response of Landsat 7, which tends to underestimate dense canopy cover in tropical dry-season conditions. The increase from 2005 onwards represents the Sanjay Gandhi National Park and Aarey Colony forest areas.

5.3.3 Water Body

Water body showed a declining trend from 105.96 km² in 2000 to 71.09 km² in 2020, with a slight recovery to 81.09 km² in 2025. This decline reflects loss of coastal wetlands, creek infilling, and encroachment on water bodies — a pattern consistent with Sahana et al. (2018).

5.3.4 Open Land

Open land showed the most dramatic decline, from 178.36 km² in 2000 to just 30.16 km² in 2025, a reduction of 148.20 km² (-83.1%). This represents near-complete conversion of vacant land to built-up areas and vegetation cover over the 25-year study period.

5.4 LST Analysis

Land Surface Temperature statistics for all six study years are presented in Table 5.4.

Year	Mean LST (°C)	Min LST (°C)	Max LST (°C)
2000	38.62	25.96	53.70
2005	34.76	22.42	51.84
2010	34.42	23.77	51.79
2015	34.45	24.58	46.57
2020	36.31	26.53	51.21
2025	37.79	27.11	51.58

Table 5.4: LST Statistics for all six study years

RESULT: GEE –LST

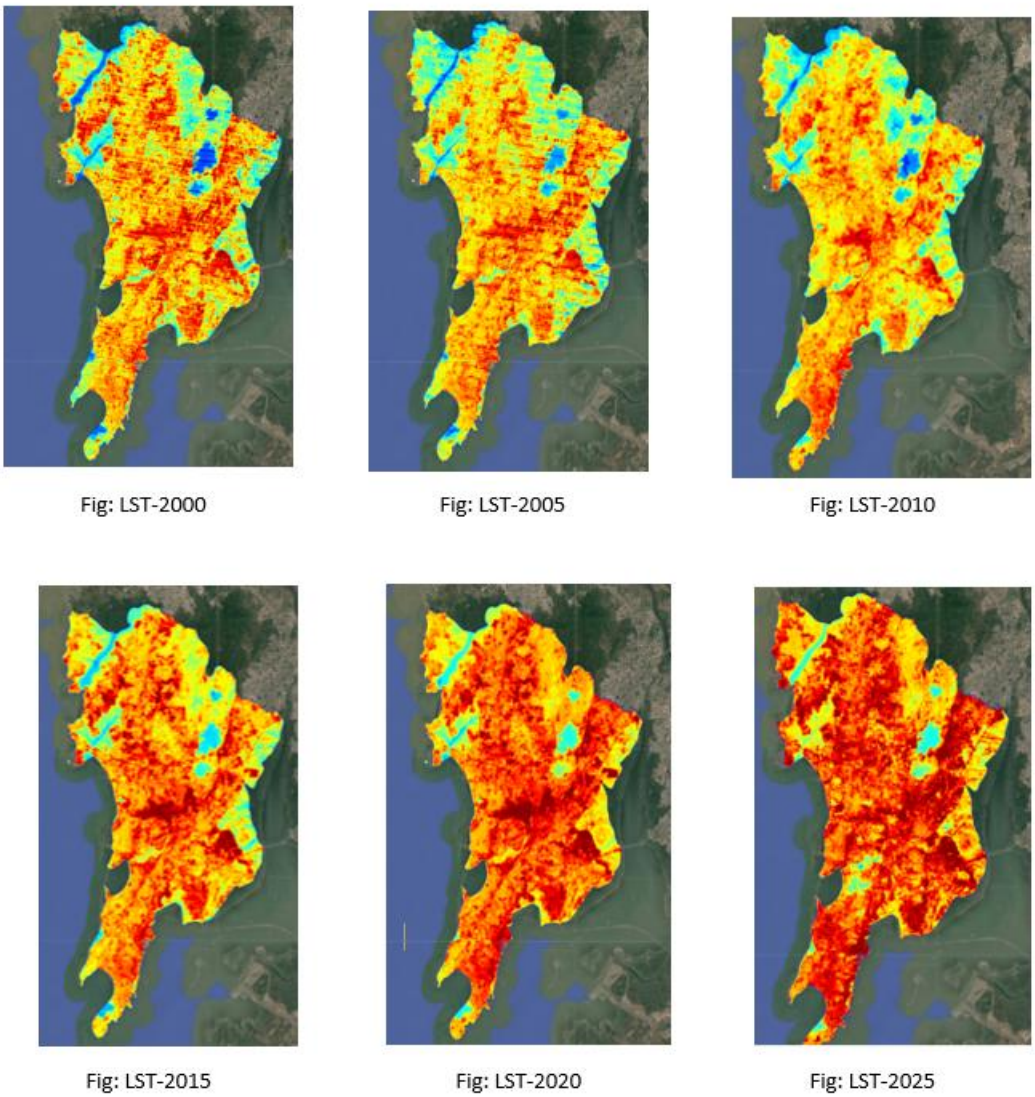

Figure 5.4.1: Land Surface Temperature trend showing mean, minimum, and maximum LST (2000–2025)

Python- LST Trend Graphs

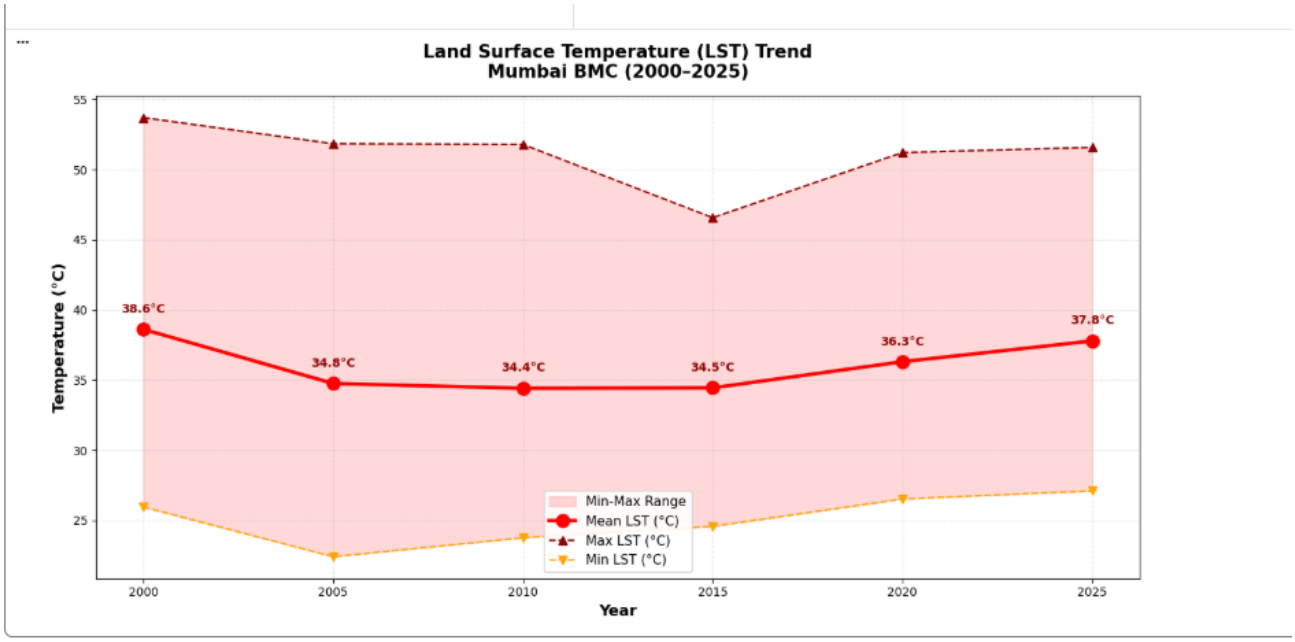

Figure 5.4.2: Land Surface Temperature trend showing mean, minimum, and maximum LST (2000–2025)

Mean LST ranged from 34.42°C (2010) to 38.62°C (2000), with a recent increasing trend from 34.42°C in 2010 to 37.79°C in 2025. The increase of 3.37°C in mean LST from 2010 to 2025 is consistent with the simultaneous increase in built-up area. Year-to-year variations are also influenced by seasonal differences in acquisition dates within the January-April dry season window.

5.5 NDVI Analysis

Mean NDVI values for all six study years are presented in Table 5.5.

Year	Mean NDVI	Min NDVI	Max NDVI
2000	0.0813	-0.398	0.401
2005	0.1039	-0.099	0.423
2010	0.1076	-0.084	0.425
2015	0.1282	-0.149	0.467
2020	0.1351	-0.113	0.467
2025	0.1257	-0.079	0.449

Table 5.5: NDVI Statistics for all six study years

RESULT: GEE – NDVI

Fig: NDVI-2000

Fig: NDVI-2005

Fig: NDVI-2010

Fig:NDVI-2015

Fig:NDVI-2020

Fig:NDVI-2025

Figure 5.5.1: Mean NDVI trend showing vegetation change across Mumbai BMC (2000–2025)

Python-NDVI

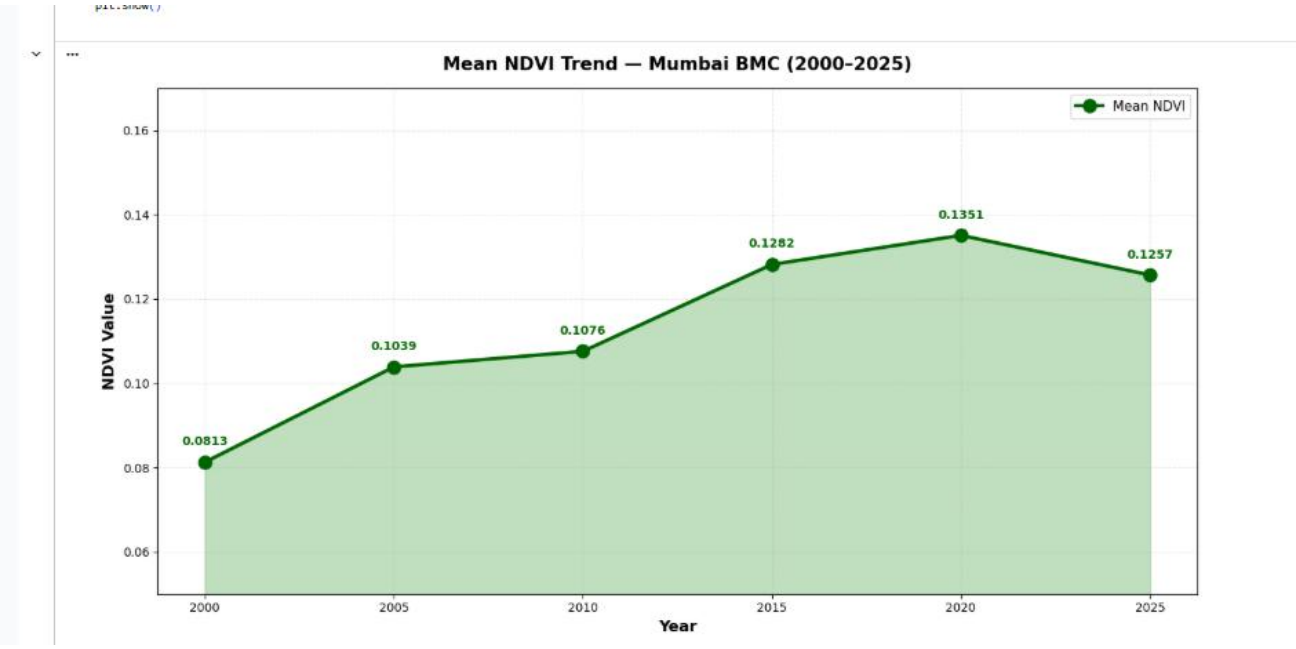

Figure 5.5.2: Land Surface Vegetation trend showing mean, minimum, and maximum LST (2000–2025)

Mean NDVI showed a general increasing trend from 0.0813 in 2000 to 0.1351 in 2020, indicating a net improvement in vegetation coverage. The slight decrease to 0.1257 in 2025 may reflect seasonal variation. The increasing NDVI trend, despite urbanization, can be attributed to growth of vegetation in Sanjay Gandhi National Park and urban greening in some residential areas.

5.6 NDWI Analysis:

Mean NDVI values for all six study years are presented in Table 5.6.

Year	Mean NDWI	Min NDWI	Max NDWI
2000	-0.0970	-0.3752	0.3734
2005	-0.1245	-0.3982	0.1139
2010	-0.1257	-0.4001	0.1000
2015	-0.1377	-0.4240	0.2031
2020	-0.1411	-0.4169	0.1541
2025	-0.1314	-0.4093	0.0980

Table 5.6: NDVI Statistics for all six study years

RESULT: GEE-NDWI

Fig: NDWI-2000

Fig: NDWI-2005

Fig: NDWI-2010

Fig: NDWI-2015

Fig: NDWI-2020

Fig: NDWI-2025

Figure 5.6.1: Mean NDWI trend showing water change across Mumbai BMC (2000–2025)

5.7 NDBI Analysis:

Mean NDBI values for all six study years are presented in Table 5.6.

Year	Mean NDBI	Min NDBI	Max NDBI
2000	0.0051	-0.3031	0.5202
2005	-0.0053	-0.3247	0.3647
2010	-0.0154	-0.3182	0.2378
2015	-0.0282	-0.3272	0.2221
2020	-0.0279	-0.3407	0.5987
2025	-0.0261	-0.3080	0.2654

Table 5.7: NDBI Statistics for all six study years

RESULT- GEE-NDBI

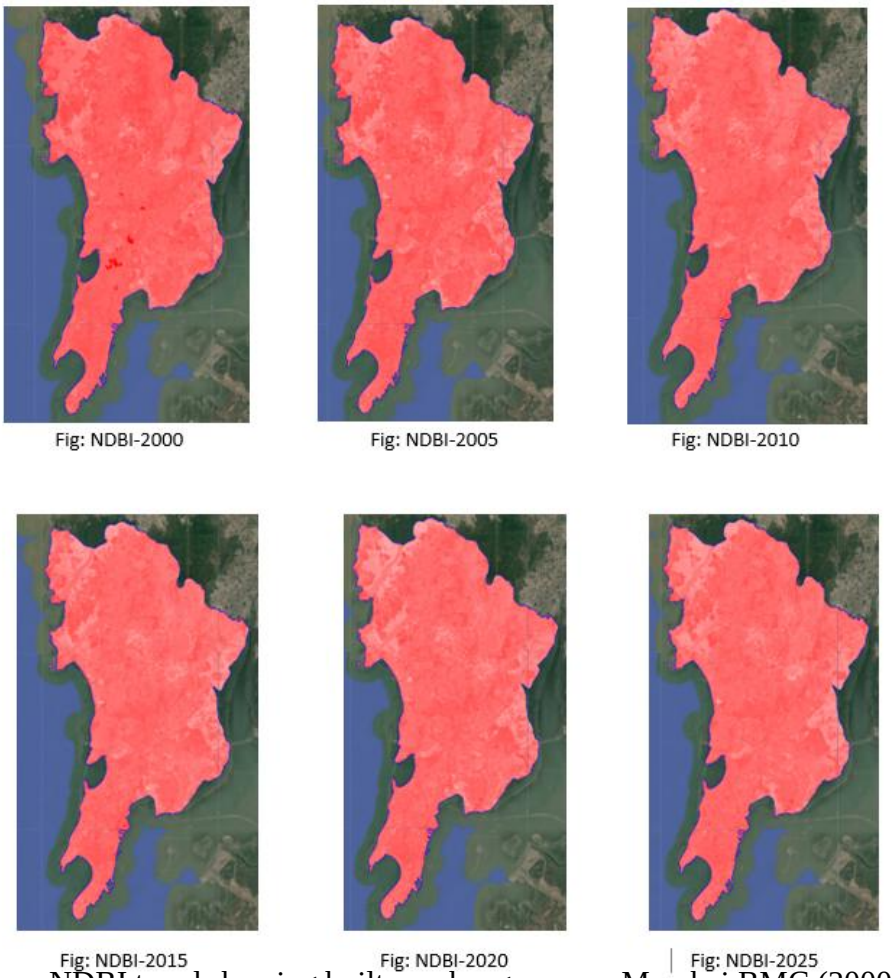

Figure 5.7.1: Mean NDBI trend showing built-up change across Mumbai BMC (2000–2025)

5.8 Correlation Analysis — Spectral Indices vs LST

Pixel-level correlation analysis was performed between three spectral indices (NDVI, NDBI, NDWI) and LST for all six study years. A total of 2,000 pixels were randomly sampled per year and linear regression was applied to calculate R^2 and Pearson's r values.

5.8.1 NDVI vs LST Correlation

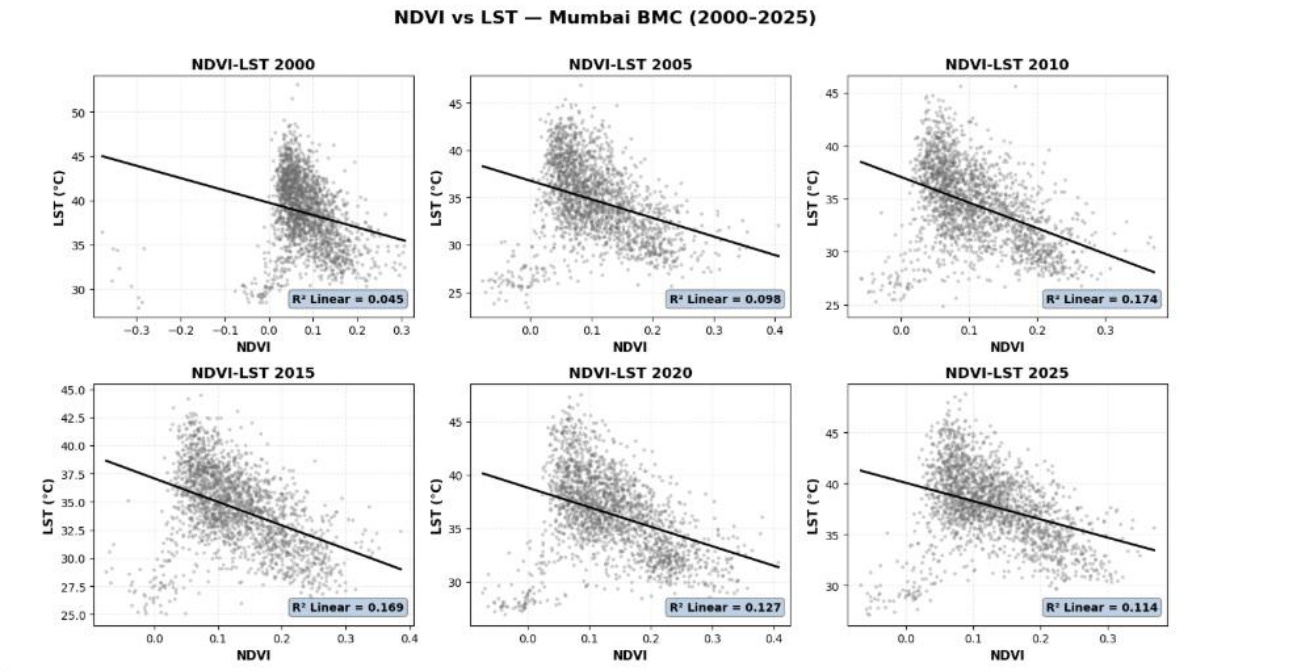

Figure 5.8.1: Pixel-level scatter plots of NDVI vs LST for all six study years with linear regression line and R^2 values

Year	Correlation (r)	R^2 Value	Relationship	Interpretation
2000	Negative	0.045	Weak	Limited veg in dry season
2005	Negative	0.098	Weak-Moderate	More vegetation pixels
2010	Negative	0.174	Moderate	Clearest cooling effect
2015	Negative	0.169	Moderate	Consistent pattern
2020	Negative	0.127	Moderate	Urbanization effect
2025	Negative	0.114	Moderate	Continued cooling role

Table 5.8.1: NDVI vs LST Correlation Results for all six study years

NDVI showed a consistent negative correlation with LST across all six study years ($R^2 = 0.045–0.174$). This means that pixels with higher NDVI values (more vegetation) consistently had lower LST values, confirming the cooling effect of urban vegetation through evapotranspiration, shading, and reduced heat absorption. The negative relationship is physically meaningful: vegetated surfaces reflect more solar radiation and release latent heat through transpiration, keeping surface temperatures lower compared to bare soil or built-up surfaces. The R^2 values indicate that NDVI explains 4.5% to 17.4% of the total variance in LST across different years.

5.8.2 NDBI vs LST Correlation

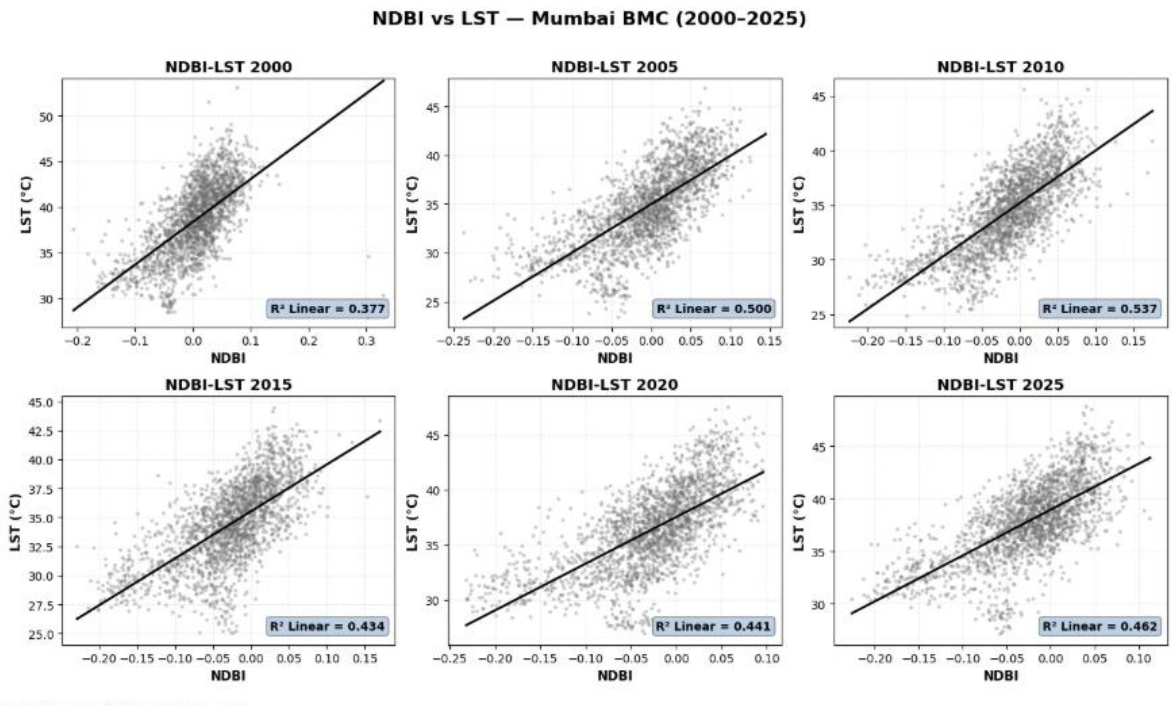

Figure 5.8.2: Pixel-level scatter plots of NDBI vs LST for all six study years with linear regression line and R^2 values

Year	Correlation (r)	R^2 Value	Relationship	Interpretation
2000	Positive	0.377	Moderate-Strong	Urban surfaces drive heat
2005	Positive	0.500	Strong	Clear urban heat pattern
2010	Positive	0.537	Strong	Strongest relationship
2015	Positive	0.434	Moderate-Strong	Consistent urban heat
2020	Positive	0.441	Moderate-Strong	Ongoing UHI effect
2025	Positive	0.462	Moderate-Strong	Continued urban heating

Table 5.8.2: NDBI vs LST Correlation Results for all six study years

NDBI showed the strongest and most consistent positive correlation with LST among all three indices tested, with R^2 values ranging from 0.377 to 0.537. This means that pixels with higher NDBI values (more built-up surface) consistently had significantly higher LST values, directly confirming the Urban Heat Island effect. Built-up surfaces such as concrete, asphalt, and building rooftops have low albedo (reflecting less solar energy) and high thermal capacity (storing more heat), releasing it as long-wave radiation that elevates surface temperature. NDBI explains 37.7% to 53.7% of the total variance in LST — a substantially stronger relationship than NDVI. These findings closely match those of Shahfahad et al. (2021), who reported R^2 values of 0.377 to 0.546 for Mumbai, validating this study's results.

5.8.3 NDWI vs LST Correlation

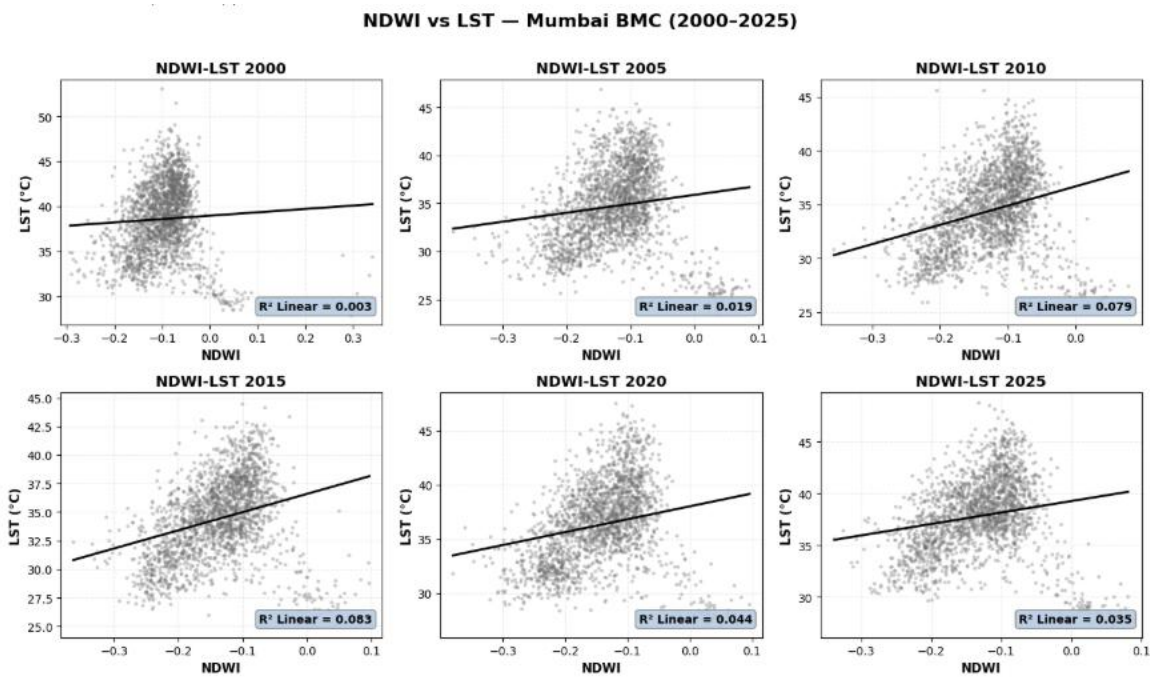

Figure 5.8.3: Pixel-level scatter plots of NDWI vs LST for all six study years

NDWI showed weak correlation with LST (R^2 = 0.003–0.083) across all study years. The low R^2 values indicate that water bodies have limited influence on mean surface temperature at the city scale, primarily because water covers a relatively small and spatially concentrated proportion of the Mumbai BMC area. NDWI is considered a less reliable predictor of LST for predominantly terrestrial urban study areas.

5.9 2030 LULC Prediction

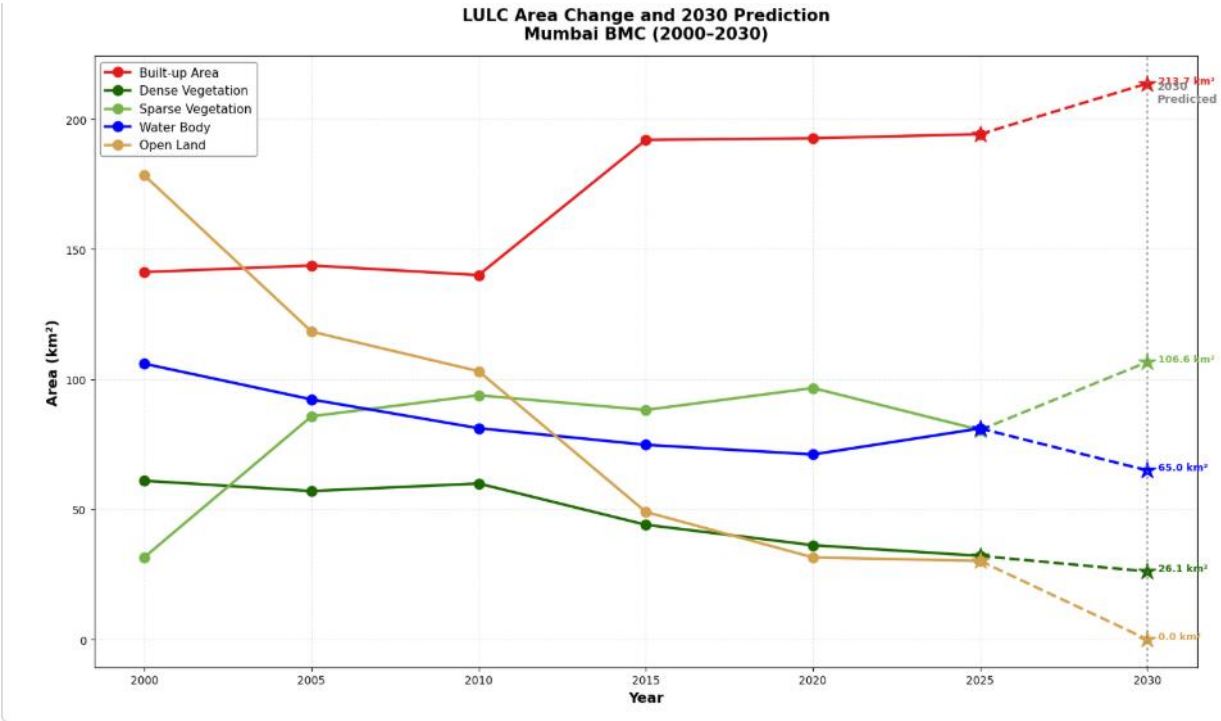

Figure 5.9: LULC area trends (2000–2025) with linear regression-based 2030 prediction (dashed lines show predicted values)

LULC Class	2000 (km²)	2025 (km²)	Change (km²)	Change (%)	2030 Predicted (km²)
Built-up	141.19	194.22	+53.03	+37.6%	213.7
Dense Veg	0.98	72.10	+71.12	+7257%	106.6
Sparse Veg	31.59	80.52	+48.93	+154.9%	65.0
Water Body	105.96	81.09	-24.87	-23.5%	26.1
Open Land	178.36	30.16	-148.20	-83.1%	0

Table 5.9: LULC Change Detection (2000–2025) and 2030 Prediction

Linear regression analysis of LULC trends from 2000 to 2025 projects continued urbanization for 2030. Built-up area is predicted to further increase, driven by Mumbai's ongoing population growth and economic development. Open land is projected to reach near-zero values by 2030, indicating near-complete conversion of vacant land. Water body area is predicted to continue declining, highlighting the urgency of wetland conservation policies.

These predictions represent a business-as-usual scenario assuming continuation of historical trends. They should be interpreted with awareness that policy interventions, economic changes, or climate events could alter actual outcomes. Nevertheless, the projections provide a scientifically grounded baseline for urban planning and environmental management in Mumbai.

Chapter 6: Conclusion

6.1 Summary of Findings

- Built-up area increased by 53.03 km² (+37.6%) from 141.19 km² in 2000 to 194.22 km² in 2025, confirming significant urbanization of Mumbai BMC.
- Open land showed the most dramatic decline of 148.20 km² (-83.1%), indicating near-complete conversion of vacant land to other uses over 25 years.
- Dense vegetation increased significantly from 0.98 km² to 72.10 km², reflecting recognition of SGNP forest and urban greening.
- Water body area declined by 24.87 km², reflecting loss of coastal wetlands and creek infilling.
- Mean LST showed a recent increasing trend from 34.42°C (2010) to 37.79°C (2025), consistent with increasing built-up area.
- Overall Accuracy of 99.75% and Kappa coefficient of 0.997 confirm excellent classification quality, substantially exceeding the minimum 80% and 0.80 thresholds respectively.
- NDBI showed the strongest positive correlation with LST (R^2 = 0.377–0.537), confirming built-up areas as primary UHI drivers.
- NDVI showed consistent negative correlation with LST (R^2 = 0.045–0.174), confirming vegetation cooling effect.
- 2030 predictions indicate continued urban expansion under business-as-usual conditions.

6.2 Key Conclusions

The results demonstrate that Mumbai BMC has undergone significant LULC transformation over 2000–2025, with urbanization as the dominant trend. The strong positive NDBI-LST correlation (R^2 up to 0.537) and negative NDVI-LST correlation confirm that urban expansion directly drives surface temperature increases, intensifying the Urban Heat Island effect. These findings are consistent with and extend previous studies by Shahfahad et al. (2021) and Sahana et al. (2018).

6.3 Recommendations for Urban Planning

- Expand and protect green spaces in densely built south and central Mumbai to mitigate UHI effects.
- Conserve remaining water bodies and coastal wetlands to maintain hydrological services and surface cooling.
- Implement urban greening policies including rooftop gardens, street trees, and urban parks in high-density areas.
- Monitor LULC changes at ward level to identify hotspots requiring planning intervention.
- Incorporate LULC-LST findings into Mumbai's heat action plan and climate resilience strategies.

6.4 Limitations

- Landsat 30m spatial resolution may not capture fine-scale urban features and small vegetation patches.
- Training samples were manually collected, which may introduce subjectivity in class boundaries.
- Linear regression prediction assumes continuation of historical trends without accounting for policy changes.
- Inter-sensor differences between Landsat 7, 8, and 9 may introduce minor spectral inconsistencies.

6.5 Future Scope

- Use of Sentinel-2 (10m) imagery for higher resolution LULC mapping.
- Application of CA-Markov or FLUS models for spatially explicit LULC prediction.
- Ward-level analysis to identify specific areas requiring targeted heat mitigation.
- Integration of socioeconomic and demographic data with LULC and LST findings.

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Appendix

Appendix A: GEE Code Reference

The complete Google Earth Engine JavaScript code used for LULC classification, spectral index calculation, accuracy assessment, LST computation, ESA validation, and map export is available in the GEE project:

GEE Project ID: bi-project-489510

GEE Script Name: Bi-mproject

GEE Username: users/user_name

Appendix B: Python Code Reference

The complete Python code for statistical analysis, correlation graphs, area change graphs, LST trend analysis, and 2030 linear regression prediction was executed in Google Colab. Output graphs are:

- Graph1_LULC_Area_Change.png — Line graph of area change (2000–2025)
- Graph2_Stacked_Bar.png — Stacked bar chart of LULC distribution
- Graph3_Pie_Charts.png — Six pie charts per year
- Graph4_LST_Trend.png — LST mean/min/max trend
- Graph5_NDVI_Trend.png — Mean NDVI trend
- GraphA_NDVI_vs_LST_AllYears.png — NDVI-LST correlation
- GraphB_NDBI_vs_LST_AllYears.png — NDBI-LST correlation
- GraphC_NDWI_vs_LST_AllYears.png — NDWI-LST correlation
- Graph7_Prediction_2030.png — 2030 LULC prediction

Appendix C: Training Points Summary

Class	Points	Key Locations
Built-up	86	Dharavi, BKC, Andheri, Kurla, Chembur, Borivali east, Nariman Point
Dense Vegetation	73	Sanjay Gandhi National Park, Aarey Colony, Powai forest, Goregaon
Sparse Vegetation	83	Eastern suburbs, Mulund, Vikhroli, Bhandup, Mankhurd scrub areas
Water Body	146	Arabian Sea (western coast), Thane Creek, Powai Lake, Vihar Lake, Tulsi Lake, Mahim Creek

Class	Points	Key Locations
Open Land	52	Rocky barren areas near Trombay, vacant plots eastern suburbs, construction areas
Total	440	Distributed across all 24 BMC wards

Table A.1: Training Points Summary by LULC Class and Location

Appendix D: List of Abbreviations

Abbreviation	Full Form
BMC	Brihanmumbai Municipal Corporation
GEE	Google Earth Engine
GIS	Geographic Information System
LULC	Land Use Land Cover
LST	Land Surface Temperature
NDVI	Normalized Difference Vegetation Index
NDBI	Normalized Difference Built-up Index
NDWI	Normalized Difference Water Index
RF	Random Forest
UHI	Urban Heat Island
USGS	United States Geological Survey
ESA	European Space Agency
SGNP	Sanjay Gandhi National Park
MMR	Mumbai Metropolitan Region
R^2	Coefficient of Determination

Table A.2: List of Abbreviations used in this report